

01:XXX:XXX - Homework n

Pranav Tikkawar

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Contents

1	L1	2
2	L2	2
3	L3	2
4	L4	2
5	L5	2
6	L6	3
7	L7	3
8	L8	4
9	L9	4
10	L10	4
11	L11	5
12	L12	6
13	L13	7
14	L14	7
15	L15	7
16	L16	8

1 L1

2 L2

3 L3

4 L4

Definition (Conditional Expectation). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, X an integrable random variable with $\mathbb{E}|X| < \infty$, and $\mathcal{G} \subseteq \mathcal{F}$ a sub- σ -algebra. A random variable Y is called the *conditional expectation* of X given $\mathcal{G}$, written

$$Y = \mathbb{E}[X \mid \mathcal{G}],$$

if

1. Y is $\mathcal{G}$ -measurable, and
2. for every $A \in \mathcal{G}$, $\int_A Y \, d\mathbb{P} = \int_A X \, d\mathbb{P}$.

When Z is a random variable, we write $\mathbb{E}[X \mid Z] := \mathbb{E}[X \mid \sigma(Z)]$.

5 L5

Definition (Filtration). A *filtration* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing sequence of sub- σ -algebras

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

We interpret $\mathcal{F}_n$ as the information available up to time n .

Definition (Martingale). Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration. A stochastic process $(Z_n)_{n \geq 0}$ is a *martingale* with respect to $(\mathcal{F}_n)$ if:

1. Z_n is $\mathcal{F}_n$ -measurable for each n (adapted),
2. $\mathbb{E}|Z_n| < \infty$ for each n , and
3. $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n$ almost surely for all $n \geq 0$.

Remark (Martingale, Submartingale, Supermartingale). With the same setup:

- (Z_n) is a *submartingale* if $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] \geq Z_n$ a.s. for all n .
- (Z_n) is a *supermartingale* if $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] \leq Z_n$ a.s. for all n .

If (Z_n) is a martingale, then $\mathbb{E}Z_n = \mathbb{E}Z_0$ for all n .

Definition (Stopping Time). Given a filtration $(\mathcal{F}_n)_{n \geq 0}$, a random variable $T : \Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$ is a *stopping time* if

$$\{T = n\} \in \mathcal{F}_n \quad \text{for every } n.$$

Equivalently, $\{T \leq n\} \in \mathcal{F}_n$ for each n .

Theorem 1 (Wald's Lemma I / Optional Stopping, Bounded Case). *Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration and $(Z_n)_{n \geq 0}$ a martingale with respect to $(\mathcal{F}_n)$. Let T be a stopping time such that $T \leq N$ almost surely for some deterministic N . Then*

$$\mathbb{E}[Z_T] = \mathbb{E}[Z_0].$$

6 L6

Definition (Almost Sure Convergence). We say $X_n \rightarrow X$ *almost surely* (a.s.) if

$$\mathbb{P}(\{\omega \in \Omega : X_n(\omega) \rightarrow X(\omega)\}) = 1.$$

Definition (Convergence in Expectation). We say $X_n \rightarrow X$ *in expectation* if $\mathbb{E}|X_n - X| \rightarrow 0$. (Often called convergence in L^1 .)

Definition (Monotone Convergence Theorem (MCT)). Let $(X_n)_{n \geq 1}$ be nonnegative random variables with $0 \leq X_1 \leq X_2 \leq \dots$ and $X_n \rightarrow X$ a.s. Then

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

Definition (Dominated Convergence Theorem (DCT)). Suppose $X_n \rightarrow X$ almost surely, and there exists an integrable random variable Y such that $|X_n| \leq Y$ a.s. for all n and $\mathbb{E}|Y| < \infty$. Then

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X].$$

7 L7

Lemma 1 (Convex Transform of a Submartingale). *Let (Z_n) be a submartingale with respect to $(\mathcal{F}_n)$ and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be convex and nondecreasing. Assume $\mathbb{E}|\varphi(Z_n)| < \infty$ for every n . Then $(\varphi(Z_n))$ is a submartingale.*

Theorem 2 (Not-To-Gamble Theorem). *Let $(X_n)_{n \geq 0}$ be a supermartingale with respect to $(\mathcal{F}_n)$, and let $(H_n)_{n \geq 1}$ be a predictable sequence (i.e. H_n is $\mathcal{F}_{n-1}$ -measurable) with $H_n \geq 0$ a.s. Define*

$$(H \cdot X)_n := \sum_{i=1}^n H_i(X_i - X_{i-1}), \quad (H \cdot X)_0 := 0.$$

Then $((H \cdot X)_n)$ is a supermartingale. In particular, no nonnegative predictable betting strategy can turn a losing game (supermartingale) into a winning game in expectation.

Theorem 3 (Optional Stopping, Bounded Case for Sub/Supermartingales). *Let $(X_n)_{n \geq 0}$ be adapted to $(\mathcal{F}_n)$ and let T be a stopping time with $T \leq N$ a.s. If (X_n) is a submartingale, then*

$$\mathbb{E}[X_N] \geq \mathbb{E}[X_T] \geq \mathbb{E}[X_0].$$

If (X_n) is a supermartingale, then

$$\mathbb{E}[X_N] \leq \mathbb{E}[X_T] \leq \mathbb{E}[X_0].$$

8 L8

Theorem 4 (Submartingale and Supermartingale Convergence). *Let $(X_n)_{n \geq 0}$ be a submartingale. If*

$$\sup_{n \geq 0} \mathbb{E}[X_n^+] < \infty,$$

then X_n converges almost surely to some integrable random variable X_∞ and $\mathbb{E}|X_\infty| < \infty$.

Let $(Z_n)_{n \geq 0}$ be a supermartingale. If there exists $c \in \mathbb{R}$ such that $Z_n \geq c$ a.s. for all n , then Z_n converges almost surely to some integrable random variable Z and $\mathbb{E}[Z] \leq \mathbb{E}[Z_0]$.

9 L9

10 L10

Theorem 5 (Markov Inequality). *Let X be a nonnegative random variable and $c > 0$. Then*

$$\mathbb{P}(X \geq c) \leq \frac{\mathbb{E}[X]}{c}.$$

Theorem 6 (Chebyshev Inequality). *Let X be a random variable with $\mathbb{E}X < \infty$ and $\text{Var}(X) < \infty$. Then for any $c > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq c) \leq \frac{\text{Var}(X)}{c^2}.$$

Theorem 7 (Kolmogorov Maximal Inequality). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}X_i = 0$ and $\text{Var}(X_i) < \infty$. Define partial sums $S_k := \sum_{i=1}^k X_i$ for $k = 1, \dots, n$, and $S_0 = 0$. Then for any $\lambda > 0$,*

$$\mathbb{P}\left(\max_{1 \leq k \leq n} |S_k| \geq \lambda\right) \leq \frac{\text{Var}(S_n)}{\lambda^2} = \frac{\sum_{i=1}^n \text{Var}(X_i)}{\lambda^2}.$$

Theorem 8 (Doob's Maximal Inequality). *Let $(M_k)_{0 \leq k \leq n}$ be a submartingale. Then for any $\lambda > 0$,*

$$\mathbb{P}\left(\max_{0 \leq k \leq n} M_k \geq \lambda\right) \leq \frac{\mathbb{E}[M_n^+]}{\lambda},$$

where $M_n^+ = \max\{M_n, 0\}$.

Theorem 9 (Ville's Inequality). *Let $(L_k)_{k \geq 0}$ be a nonnegative supermartingale with respect to $(\mathcal{F}_k)$. Then for any $x > 0$,*

$$\mathbb{P}\left(\sup_{k \geq 0} L_k \geq x\right) \leq \frac{\mathbb{E}[L_0]}{x}.$$

Theorem 10 (Fatou's Lemma). *Let $(X_n)_{n \geq 1}$ be nonnegative random variables. Then*

$$\mathbb{E}\left[\liminf_{n \rightarrow \infty} X_n\right] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n].$$

11 L11

Definition (σ^2 -Subgaussian Random Variable). A random variable X is called σ^2 -subgaussian if $\mathbb{E}X = 0$ and

$$\mathbb{E}[e^{sX}] \leq \exp\left(\frac{\sigma^2 s^2}{2}\right) \quad \text{for all } s \in \mathbb{R}.$$

Lemma 2 (Hoeffding's Lemma and Hoeffding's Inequality). *If X satisfies $a \leq X \leq b$ a.s. and $\mathbb{E}X = 0$, then for all $s \in \mathbb{R}$,*

$$\mathbb{E}[e^{sX}] \leq \exp\left(\frac{s^2(b-a)^2}{8}\right),$$

so X is $(b-a)^2/4$ -subgaussian.

If $X_1, \dots, X_n$ are independent random variables with $a_i \leq X_i \leq b_i$ a.s., and $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$, then for any $t > 0$,

$$\mathbb{P}(\bar{X}_n - \mathbb{E}\bar{X}_n \geq t) \leq \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and similarly for the two-sided deviation.

Theorem 11 (Azuma-Hoeffding Inequality for Martingales). *Let $(Z_n)_{n \geq 0}$ be an integrable martingale with respect to $(\mathcal{F}_n)$, and assume there exist constants $\alpha_i, \beta_i \geq 0$ such that*

$$-\alpha_i \leq Z_i - Z_{i-1} \leq \beta_i \quad \text{a.s. for all } i \geq 1.$$

Then for any $n \geq 1$ and $t > 0$,

$$\mathbb{P}(Z_n - \mathbb{E}Z_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (\alpha_i + \beta_i)^2}\right),$$

and the same bound holds for $\mathbb{P}(|Z_n - \mathbb{E}Z_n| \geq t)$ up to a factor 2.

Theorem 12 (McDiarmid's Inequality). *Let $Z = h(X_1, \dots, X_n)$ where $X_1, \dots, X_n$ are independent, and suppose h satisfies the bounded difference condition: whenever x and $\tilde{x}$ differ only in coordinate i , we have*

$$|h(x) - h(\tilde{x})| \leq s.$$

Then for any $t > 0$,

$$\mathbb{P}(Z - \mathbb{E}Z \geq t) \leq \exp\left(-\frac{2t^2}{s^2 n}\right),$$

and similarly for the two-sided deviation with an extra factor 2.

12 L12

Definition (Transition Probability Matrix). For a time-homogeneous Markov chain (X_n) on a finite state space $S = \{x_1, \dots, x_m\}$, the *transition matrix* $P \in \mathbb{R}^{m \times m}$ is defined by

$$P(i, j) := \mathbb{P}(X_{n+1} = x_j \mid X_n = x_i),$$

which is independent of n . Each row of P is a probability vector.

Example (Biased Random Walk with absorbing Boundary).

$$h(i) = \begin{cases} \frac{1-(q/p)^i}{1-(q/p)^N} & \text{if } p \neq q, \\ \frac{i}{N} & \text{if } p = q, \end{cases}$$

where $h(i) = \mathbb{P}(\text{hit } N \text{ before } 0 \mid X_0 = i)$.

Theorem 13 (Chapman–Kolmogorov Equation (Finite Case)). *For a time-homogeneous Markov chain with transition matrix P , we have*

$$\mathbb{P}(X_{n+k} = x_j \mid X_n = x_i) = (P^k)(i, j)$$

for all $n, k \geq 0$ and all states $x_i, x_j \in S$.

Definition (Stationary Distribution). A probability vector π on S is called a *stationary distribution* for a Markov chain with transition matrix P if

$$\pi = \pi P.$$

Equivalently, if $X_0 \sim \pi$, then $X_n \sim \pi$ for all $n \geq 0$.

Definition (Detailed Balance and Reversible Chains). A distribution π on S satisfies the *detailed balance* equations with respect to P if

$$\pi(i)P(i, j) = \pi(j)P(j, i) \quad \text{for all } i, j.$$

In this case the chain is called *reversible* with respect to π , and π is automatically stationary.

Definition (Metropolis–Hastings Algorithm). fix a distribution $q(x, \cdot)$ on S for each $x \in S$. Given a target distribution π on S , we define a Markov chain with transition matrix.

$$P(x, y) = q(x, y)\alpha(x, y) \quad \text{for } y \neq x, \quad P(x, x) = 1 - \sum_{y \neq x} P(x, y),$$

where

$$\alpha(x, y) = \min \left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)} \right).$$

Set $X_{n+1} = \begin{cases} y & \text{with probability } P(X_n, y), \\ X_n & \text{with probability } P(X_n, X_n). \end{cases}$ Then π is a stationary distribution for this chain.

Definition (Gibbs Sampler). Let π be a distribution on $S = S_1 \times \cdots \times S_d$. For each i , define the conditional distribution

$$\pi_i(x_i \mid x_{-i}) := \mathbb{P}_{X \sim \pi}(X_i = x_i \mid X_j = x_j \text{ for } j \neq i).$$

The Gibbs sampler is a Markov chain on S that updates one coordinate at a time in a random order. We choose i uniformly at random from $\{1, \dots, d\}$, and then update X_i according to $\pi_i(\cdot \mid X_{-i})$. This chain has π as a stationary distribution.

13 L13

14 L14

Remark (Extended Stationary Distribution Properties). For a finite-state Markov chain with transition matrix P :

- A stationary distribution always exists.
- If the chain is irreducible, the stationary distribution is unique.
- If the chain is irreducible and aperiodic, the distribution of X_n converges to the stationary distribution as $n \rightarrow \infty$ (you can cite this without proof).

Irreducible means every state can reach every other in finitely many steps. Aperiodicity rules out deterministic cycles such as a two-state alternation.

15 L15

Definition (Irreducibility and Aperiodicity). A Markov chain with transition matrix P is *irreducible* if for every pair of states $x_i, x_j \in S$, there exists $k \geq 0$ such that $(P^k)(i, j) > 0$. The chain is *aperiodic* if for every state x_i , the greatest common divisor of $\{k \geq 1 : (P^k)(i, i) > 0\}$ is 1.

Definition (Period). The *period* of a state x_i is the greatest common divisor of $\{k \geq 1 : (P^k)(i, i) > 0\}$. If the period is greater than 1, the chain exhibits deterministic cycles and does not converge to a stationary distribution.

Theorem 14 (Markov Chain Convergence). *If a Markov chain with transition matrix P is irreducible and aperiodic, then there exists a unique stationary distribution π , and for any initial distribution μ ,*

$$\lim_{n \rightarrow \infty} \mu P^n = \pi.$$

Furthermore, the convergence is geometric in total variation distance, i.e. there exist constants $C > 0$ and $\rho \in (0, 1)$ such that for all n ,

$$\|\mu P^n - \pi\|_{\text{TV}} \leq C \rho^n.$$

Where $\|\cdot\|_{\text{TV}}$ denotes the total variation distance between probability distributions.

$$\|p - q\|_{\text{TV}} = \frac{1}{2} \sum_{x \in S} |p(x) - q(x)|$$

16 L16

Definition (Reversible Markov Chain). A Markov chain with transition matrix P is *reversible* with respect to a distribution π if π satisfies the detailed balance equations

$$\pi(i)P(i, j) = \pi(j)P(j, i) \quad \text{for all } i, j.$$

Definition (Spectral Decomposition of Reversible Markov Chains). If a Markov chain is reversible with respect to π , then P is self-adjoint on the Hilbert space $L^2(\pi)$, and admits an orthonormal basis of eigenvectors $\{f_k\}$ with corresponding real eigenvalues $\{\lambda_k\}$. The largest eigenvalue is $\lambda_1 = 1$ with eigenvector $f_1 \equiv 1$, and all other eigenvalues satisfy $|\lambda_k| < 1$.